Machine Psychology: One-Page Research Summary

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WHAT MACHINE PSYCHOLOGY IS

Machine Psychology treats large language models as behavioural subjects. Instead of asking only whether a model is accurate or aligned on benchmarks, it runs controlled, case-based behavioural experiments, adapted from clinical and differential psychology, that map *how* a model's behaviour changes with who it believes it is talking to, what persona it is given, and what pressures the interaction applies. The programme is independent: 15 completed studies since 2025 (three co-authored with M. Borhani), all 15 posted as preprints (twelve on Research Square, three on Zenodo), and all 15 manuscripts under review at journals or a workshop, or in active submission.

THREE HEADLINE FINDINGS

1. **Identity-reactive behaviour.** In the flagship study (*Ten Novel Phenomena in Machine Psychology*, under review at *AI and Ethics*, Springer Nature), models systematically shift behaviour in response to ethnically cued user names. The effect is invisible to standard capability benchmarks and directly relevant to fairness in deployment.

2. **Reality distortion under pressure.** *Algorithmic Gaslighting* shows models can be induced, through structured conversational pressure, to deny or rewrite their own earlier outputs; the study proposes a provisional Reality Distortion Index (RDI) as a first measurement heuristic.

3. **Performative self-correction.** *When AI Reviews Itself* finds zero answer changes across 72 self-correction rounds, suggesting that visible "self-critique" can be a surface ritual rather than a genuine revision process, with consequences for how self-evaluation is used in alignment pipelines.

Further studies map deception awareness (*The Lie Awareness Spectrum*), self-censorship and metacognitive access (*Thinking Behind the Mask*), persona-dependent behavioural fracturing (*Ontological Firewalls*), and moral judgement under role-play (*Moral Inversion*).

WHERE I WANT TO TAKE THIS IN A PHD

- From case studies to **systematic behavioural phenotyping**: preregistered, cross-model behavioural batteries with the controls the field currently lacks.
- From phenomena to **mechanisms**: pairing behavioural results with interpretability methods to test why identity and persona cues move behaviour.
- From description to **evaluation practice**: behavioural test suites that labs and auditors can run before deployment, alongside capability benchmarks.

FIT WITH A RESEARCH GROUP

What I bring is the phenomena side: a set of documented effects, the protocols that produced them, and fifteen finished studies. The PhD is where I want to learn to do this more rigorously, moving from case studies toward preregistered, cross-model behavioural batteries that evaluation and alignment teams can actually use.

SELECTED LINKS

- Flagship preprint: doi.org/10.21203/rs.3.rs-10369594/v1
- Google Scholar: https://scholar.google.com/citations?user=FWqAHkgAAAAJ
- Full publication list: attached CV